

## 6.5 Escape-sequences

What is an escape-sequence?

An escape-sequence is just a name for a code that is sent to the Printer from the computer to change the parameters that control the way the Printer works. Many parameters that can be set via the operator panel, can also be changed with escape-sequences. Note that the escape-sequences will be active on the print job they are enclosed, but the Printer will go back to what the operator panel dictates when finished with the job.

This makes it possible to have different codes (read; escape-sequences) stored in a document. So, when the document is sent to the Printer, these codes are sent first and the Printer sets itself in the correct mode and format automatically.



Note! When the Printer receives an escape-sequence, it will have first priority. This means, regardless of the setting made before and regardless what the operator panel dictates, the most recent escape-sequence will take precedence.



Note! Escape-sequences should be placed at the very beginning of the first page on the sheet, (e.g. page 1, 3, 5, 7 ...). However, a software form feed can be placed wherever needed. (If there is a command on the other pages, it will be skipped).  
All page formatting which is done on the front page will also format the back of the same sheet.

An overview of the different escape-sequences:

|          |                                                       |                                               |
|----------|-------------------------------------------------------|-----------------------------------------------|
| ESC A n  | - Sheet Width.                                        | n can be 0 (2-page mode) or 1 (4-page mode).  |
| ESC B nn | - Line length.                                        | nn can be from 10 to 44 characters.           |
| ESC D nn | - Margin 1.                                           | nn can be from 0.5" to 2.0"                   |
| ESC E nn | - Margin 2.                                           | nn can be from 0.5" to 2.0"                   |
| ESC G n  | - Page 1 up or down                                   | n can be 0 (Page 1 up) or 1 (Page 1 down).    |
| ESC H n  | - Print format                                        | n can be 0 (Single sided) or 1 (Inter point). |
| ESC P nn | - No. of sheets                                       | nn can be from 01 to 99 sheets                |
| ESC 0    | - Soft Reset                                          |                                               |
| ESC 1    | - Soft Form Feed                                      |                                               |
| ESC 4    | - Adding the Printer ID at the last line on the sheet |                                               |



Note! The escape-sequences will be executed immediately if the Printer is not running. However, if the Printer is running, the escape-sequences will keep their place in the document, and will be executed when this particular page is printed. Also keep in mind that any page formatting command must be kept on the front page of a sheet. Then the command will affect both the front and back page, i.e. one sheet. If there are page formatting commands on the back page of a sheet, these commands will be skipped.

Please see the following explanations on how to combine different values to get the different escape-sequences.

### Sheet Width:

ESC A n - Sheet width.

n can be 0 or 1, 2-page mode (0) or 4-page mode (1).

| Page mode | ASCII       | HEX      |
|-----------|-------------|----------|
| 2-page    | 027 065 048 | 1B 41 30 |
| 4-page    | 027 065 049 | 1B 41 31 |

Default is 4-page.

### Line length:

ESC B nn - Line length.

nn can be from 10 to 44 characters.

| Char | ASCII           | HEX         |
|------|-----------------|-------------|
| 10   | 027 066 049 048 | 1B 42 31 30 |
| 11   | 027 066 049 049 | 1B 42 31 31 |
| 12   | 027 066 049 050 | 1B 42 31 32 |
| 13   | 027 066 049 051 | 1B 42 31 33 |
| 14   | 027 066 049 052 | 1B 42 31 34 |
| 15   | 027 066 049 053 | 1B 42 31 35 |
| 16   | 027 066 049 054 | 1B 42 31 36 |
| 17   | 027 066 049 055 | 1B 42 31 37 |
| 18   | 027 066 049 056 | 1B 42 31 38 |
| 19   | 027 066 049 057 | 1B 42 31 39 |
| 20   | 027 066 050 048 | 1B 42 32 30 |
| 21   | 027 066 050 049 | 1B 42 32 31 |
| 22   | 027 066 050 050 | 1B 42 32 32 |
| 23   | 027 066 050 051 | 1B 42 32 33 |
| 24   | 027 066 050 052 | 1B 42 32 34 |
| 25   | 027 066 050 053 | 1B 42 32 35 |
| 26   | 027 066 050 054 | 1B 42 32 36 |
| 27   | 027 066 050 055 | 1B 42 32 37 |
| 28   | 027 066 050 056 | 1B 42 32 38 |
| 29   | 027 066 050 057 | 1B 42 32 39 |
| 30   | 027 066 051 048 | 1B 42 33 30 |
| 31   | 027 066 051 049 | 1B 42 33 31 |
| 32   | 027 066 051 050 | 1B 42 33 32 |
| 33   | 027 066 051 051 | 1B 42 33 33 |

|    |                 |             |
|----|-----------------|-------------|
| 34 | 027 066 051 052 | 1B 42 33 34 |
| 35 | 027 066 051 053 | 1B 42 33 35 |
| 36 | 027 066 051 054 | 1B 42 33 36 |
| 37 | 027 066 051 055 | 1B 42 33 37 |
| 38 | 027 066 051 056 | 1B 42 33 38 |
| 39 | 027 066 051 057 | 1B 42 33 39 |
| 40 | 027 066 052 048 | 1B 42 34 30 |
| 41 | 027 066 052 049 | 1B 42 34 31 |
| 42 | 027 066 052 050 | 1B 42 34 32 |
| 43 | 027 066 052 051 | 1B 42 34 33 |
| 44 | 027 066 052 052 | 1B 42 34 34 |

Default is 42 characters per. line.

### Margin 1:

ESC D nn - Margin 1.

nn can be from 0.5" to 2.0"

| Margin 1 | ASCII           | HEX         |
|----------|-----------------|-------------|
| 0.5"     | 027 068 048 053 | 1B 44 30 35 |
| 0.6"     | 027 068 048 054 | 1B 44 30 36 |
| 0.7"     | 027 068 048 055 | 1B 44 30 37 |
| 0.8"     | 027 068 048 056 | 1B 44 30 38 |
| 0.9"     | 027 068 048 057 | 1B 44 30 39 |
| 1.0"     | 027 068 049 048 | 1B 44 31 30 |
| 1.1"     | 027 068 049 049 | 1B 44 31 31 |
| 1.2"     | 027 068 049 050 | 1B 44 31 32 |
| 1.3"     | 027 068 049 051 | 1B 44 31 33 |
| 1.4"     | 027 068 049 052 | 1B 44 31 34 |
| 1.5"     | 027 068 049 053 | 1B 44 31 35 |
| 1.6"     | 027 068 049 054 | 1B 44 31 36 |
| 1.7"     | 027 068 049 055 | 1B 44 31 37 |
| 1.8"     | 027 068 049 056 | 1B 44 31 38 |
| 1.9"     | 027 068 049 057 | 1B 44 31 39 |
| 2.0"     | 027 068 050 048 | 1B 44 32 30 |

Default is 1.0".

### Margin 2:

ESC E nn - Margin 2.  
nn can be from 0.5" to 2.0"

| Margin 2 | ASCII           | HEX         |
|----------|-----------------|-------------|
| 0.5"     | 027 069 048 053 | 1B 45 30 35 |
| 0.6"     | 027 069 048 054 | 1B 45 30 36 |
| 0.7"     | 027 069 048 055 | 1B 45 30 37 |
| 0.8"     | 027 069 048 056 | 1B 45 30 38 |
| 0.9"     | 027 069 048 057 | 1B 45 30 39 |
| 1.0"     | 027 069 049 048 | 1B 45 31 30 |
| 1.1"     | 027 069 049 049 | 1B 45 31 31 |
| 1.2"     | 027 069 049 050 | 1B 45 31 32 |
| 1.3"     | 027 069 049 051 | 1B 45 31 33 |
| 1.4"     | 027 069 049 052 | 1B 45 31 34 |
| 1.5"     | 027 069 049 053 | 1B 45 31 35 |
| 1.6"     | 027 069 049 054 | 1B 45 31 36 |
| 1.7"     | 027 069 049 055 | 1B 45 31 37 |
| 1.8"     | 027 069 049 056 | 1B 45 31 38 |
| 1.9"     | 027 069 049 057 | 1B 45 31 39 |
| 2.0"     | 027 069 050 048 | 1B 45 32 30 |

Default is 1.0".

### Page 1 up or down:

ESC G n - Page 1 up/down.  
n can be 0 (up) or 1 (down).

| Page 1 | ASCII       | HEX      |
|--------|-------------|----------|
| Up     | 027 071 048 | 1B 47 30 |
| Down   | 027 071 049 | 1B 47 31 |

Default is Page 1 Up.

### Print format:

ESC H n - Print format  
n can be 0 (single sided) or 1 (inter point).

| Print format | ASCII       | HEX      |
|--------------|-------------|----------|
| Single sided | 027 072 048 | 1B 48 30 |
| Inter point  | 027 072 049 | 1B 48 31 |

Default is inter point.

### Number of sheets:

ESC P nn - Number of sheets  
nn can be 01 to 99 sheets.

| Sheets | ASCII           | HEX         |
|--------|-----------------|-------------|
| 01     | 027 080 048 049 | 1B 50 30 31 |
| 02     | 027 080 048 050 | 1B 50 30 32 |
| 03     | 027 080 048 051 | 1B 50 30 33 |
| 04     | 027 080 048 052 | 1B 50 30 34 |
| 05     | 027 080 048 053 | 1B 50 30 35 |
| 06     | 027 080 048 054 | 1B 50 30 36 |
| 07     | 027 080 048 055 | 1B 50 30 37 |
| 08     | 027 080 048 056 | 1B 50 30 38 |
| 09     | 027 080 048 057 | 1B 50 30 39 |
| 10     | 027 080 049 048 | 1B 50 31 30 |
| 11     | 027 080 049 049 | 1B 50 31 31 |
| 12     | 027 080 049 050 | 1B 50 31 32 |
| 13     | 027 080 049 051 | 1B 50 31 33 |
| 14     | 027 080 049 052 | 1B 50 31 34 |
| 15     | 027 080 049 053 | 1B 50 31 35 |
| 16     | 027 080 049 054 | 1B 50 31 36 |
| 17     | 027 080 049 055 | 1B 50 31 37 |
| 18     | 027 080 049 056 | 1B 50 31 38 |
| 19     | 027 080 049 057 | 1B 50 31 39 |
| 20     | 027 080 050 048 | 1B 50 32 30 |

### Software Reset:

ESC 0 - Soft Reset

This command is used to reset the Printer. It is used from the computer and has the same effect as pushing the key RESET PRINTER. Software Reset should be used with care: If the printer has not finished printing, the rest of the text in the buffer will be lost, and a new paper position will be assumed by the Printer. Because of this, the command is only to be used after a software form feed has been executed, and the Printer has stopped completely.

|                | ASCII   | HEX   |
|----------------|---------|-------|
| Software Reset | 027 048 | 1B 30 |

### Software Form Feed:

ESC 1 - Soft Form Feed

This command is to be used after all text in one volume has been transmitted to the Printer. If text corresponding to less than two pages, or text with an odd number of pages is received, and not followed by FF on the last page, the Printer will wait for more text or FF. This means that the last page may be stuck in the Printer. This is due to the double-sided printing of the Printer. This command makes the Printer to start printing the rest of the text. After this the paper position will be the same as it had when this volume of text was started. Then page no. 1 on the next volume will start out correctly. There will always be at least one blank sheet of paper between the volumes of text when finishing each volume with a Software Form Feed.

|                    | ASCII   | HEX   |
|--------------------|---------|-------|
| Software Form Feed | 027 049 | 1B 31 |

### Printer ID:

ESC 4 - Adding the Printer ID

When you have several Braille Printers producing the same Braille material, it is not always easy to later figure out what Printer made which book. It is therefore possible to add a Printer ID to the last line on a sheet. There is no matter where the escape-sequence is placed on the page, the ID will always be printed on the last line. Meaning, if you send this escape-sequence on the last page of every print job, you will get the Printer ID printed on the last line on the last page.

The Printer ID is a 5 digit code that has to be set in the menu choice: *Main menu - Printer Setup - Printer ID*. See figure below:

